Curriculum-Guided Siamese Real-ESRGAN with High-Order Degradation for Real-World Image Super-Resolution

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Abstract. Real-world image degradation combines blur, sensor noise, chromatic aberration, and JPEG compression, making synthetic restoration training difficult. We present a Siamese Real-ESRGAN framework that improves degradation diversity and training stability for realistic image restoration. The method extends high-order degradation with probabilistic operation selection, tuned compression ranges, camera-noise simulation, and offline multi-level quality generation. A shared-weight Teacher–Student training scheme restores paired inputs with different degradation severity, while a three-stage curriculum gradually shifts from mild to severe degradation pairs. Experiments on standard and real-world benchmarks show improved perceptual quality, reaching NIQE 3.82 on Urban100 and an 18.7% NIQE gain over Real-ESRGAN on RealSR Canon, without increasing inference cost.

Keywords: Image Super-Resolution · Siamese Network · Knowledge Distillation · Realistic Degradation · Real-ESRGAN

1 Introduction

Single image super-resolution (SR) aims to recover a high-resolution image from a degraded low-resolution input. It plays an important role in various domains including digital photography, surveillance, remote sensing, medical imaging, and visual content enhancement, where image details are often lost due to limited sensor resolution, compression, noise, blur, or transmission constraints. Although recent deep learning-based SR methods have achieved remarkable results, many of them are trained under simplified synthetic degradations which do not fully represent real-world image degradation.

In practical scenarios, low-quality images usually contain a complex mixture of unknown degradations, including optical blur, sensor noise, JPEG compression, resizing artifacts, and camera post-processing effects. This gap between synthetic training data and real-world inputs often causes SR models to generate over-smoothed textures, amplified noise, ringing artifacts, or visually unrealistic details. To address this issue, real-world SR methods such as ESRGAN [**?**], BSRGAN [**?**], and Real-ESRGAN [**?**] introduce more realistic degradation models and perceptual learning objectives. Among them, Real-ESRGAN has become a strong baseline by using a high-order degradation pipeline to simulate complex real-world artifacts.

However, learning from real-world degradations remains challenging because image quality may vary from mildly degraded to severely corrupted. Training directly on strong degradations can make optimization unstable and may lead to unreliable texture generation, while training mainly on mild degradations may reduce robustness on difficult inputs. Inspired by curriculum learning [?], a natural strategy is to expose the model to multiple degradation levels in a progressive manner, allowing it to first learn from easier cases where structural information is still preserved and then gradually adapt to harder cases with stronger artifacts. In this work, we adopt this mild-to-hard training strategy by constructing paired degradation levels from the same high-resolution image, so that easier degradation cases can support the learning of more difficult restoration scenarios.

The main contributions of this paper are two-fold. First, we propose a shared-weight Siamese Real-ESRGAN training scheme that improves restoration guidance without increasing inference-time complexity. Second, we design a mild-to-hard curriculum scheme using progressive degradation pairs and an enhanced high-order degradation pipeline to stabilize learning under increasingly severe degradations.

2 Related Work

This section reviews representative studies related to image super-resolution, with emphasis on real-world degradation modeling and knowledge distillation.

Deep learning has significantly improved image super-resolution by learning the mapping from low-quality to high-quality images. Early CNN-based methods such as SRCNN [?] demonstrated the effectiveness of end-to-end learning, while deeper residual and dense models such as EDSR [?] and RDN [?] further improved reconstruction accuracy. To enhance perceptual quality, ESRGAN [?] introduced adversarial learning and perceptual loss, generating sharper and more realistic textures. More recent Transformer-based models, such as SwinIR [?], have also shown strong restoration performance. However, many of these methods are still trained with simplified degradations, which limits their robustness in real-world scenarios.

To address the mismatch between synthetic training data and practical inputs, recent studies have paid increasing attention to degradation modeling. Low-quality images captured in practice often contain mixed and unknown degradations, including blur, noise, compression, resizing artifacts, and camera processing effects. Blind SR methods attempt to restore images without assuming a fixed degradation model. BSRGAN [?] improves degradation synthesis by combining blur, downsampling, noise, and JPEG compression in a more practical manner. Real-ESRGAN [?] further introduces a high-order degradation process to simulate complex real-world artifacts and has become a strong baseline for practical SR. Our work builds on this direction, but focuses on exploiting different degradation severity levels during training.

Another line of related research is knowledge transfer for restoration networks. Knowledge distillation is commonly used to transfer information from a Teacher model to a Student model [?]. In SR, several distillation methods aim to improve lightweight models by transferring feature-level or output-level knowledge. However, most of them rely on separate Teacher and Student networks or mainly target model compression. Different from these approaches, our method uses two degraded versions of the same image and a shared generator, allowing the milder degradation level to guide the harder one.

This design is also related to curriculum learning [?], where models are trained from easier to harder samples to improve optimization stability. In our setting, difficulty is defined by degradation strength. The model is first guided by relatively mild degradation pairs and then gradually exposed to more severe degradation pairs. By adopting this mild-to-hard degradation curriculum, our method aims to stabilize training and improve robustness under increasingly difficult real-world degradations.

3 Methodology

The proposed framework has two components: multi-level realistic degradation and weight-sharing Siamese training. Fig. 1 summarizes the full pipeline, which builds on high-order degradation and organizes it into paired severity levels for curriculum learning.

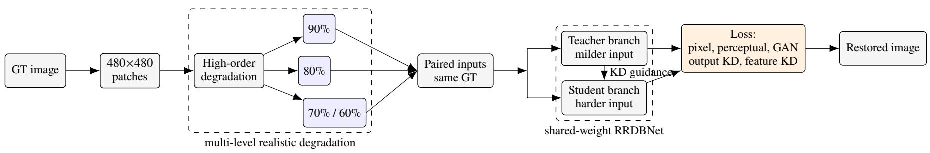

Fig. 1. Overview of the proposed Siamese Real-ESRGAN framework. Multi-level degraded samples are paired by curriculum stage and processed by one shared-weight generator in two passes.

3.1 Multi-Level High-Order Degradation

Based on the high-order degradation pipeline of Real-ESRGAN and the refined setting in [?], we retain the two-stage structure while adding probabilistic operation selection, practical JPEG ranges, and camera-specific noise, including chromatic aberration and sensor shot/read noise. For an HQ image I_{HQ}, the generated LQ image at quality level q is defined as

$$I_{LQ}^{(q)} = \mathcal{D}_q(I_{HQ}) = \Pi\big(R_{0.25}, B_{\Theta_q}, N_{\Theta_q}, C, J_{\Theta_q}\big)(I_{HQ}),$$

where R, B, N, J, and C denote resizing, blur, noise, JPEG compression, and camera-specific noise, respectively. The resize factor is fixed to 0.25 for $\times 4$ SR. In $\Pi(\cdot)$, resizing is applied first in the first pass, while blur, noise, camera noise, and JPEG compression are sampled probabilistically; the second pass then applies the selected non-resize degradations. The level $q \in \{90, 80, 70, 60\}$ is a quality index: $q = 100$ denotes the original GT image, larger q indicates milder degradation, and lower q indicates stronger degradation.

Each quality level changes the parameter set

$$\Theta_q = (\Sigma_q, Q_q, p_q^B),$$

Realdegradationstructure.png

Fig. 2. Enhanced two-stage degradation pipeline for multi-level LQ generation.

where Σ_q is the Gaussian-noise sigma range, Q_q is the JPEG quality range, and p_q^B is the blur probability:

q	Σ_q	Q_q	p_q^B
90	$[1, 10]$	$[75, 95]$	0.4
80	$[5, 15]$	$[60, 85]$	0.6
70	$[10, 20]$	$[45, 70]$	0.7
60	$[15, 25]$	$[30, 55]$	0.8

The remaining probabilities are shared across levels: $p^{resize} = 1.0$, $p^{noise} = 0.9$, $p^{JPEG} = 0.9$, and $p^{camera} = 0.7$. Therefore, 90% samples preserve more structure with weaker noise, lighter compression, and less frequent blur, while 60% samples contain stronger noise, heavier JPEG artifacts, and more frequent blur.

For $\times 4$ SR, GT images are cropped into 480×480 patches with 50% overlap, producing 120×120 LQ counterparts. Each patch is pre-degraded into four quality levels, enabling efficient and reproducible 90/80, 80/70, and 70/60 pairing.

3.2 Siamese Training with Shared Weights

Fig. **??** details how the paired degradation levels are processed by the shared-weight Teacher–Student branches during training. The generator is the Real-ESRGAN RRDB-Net backbone [?,?] with 23 RRDB blocks and 64 feature channels. Each mini-batch contains two degraded versions of the same GT image. The milder Teacher pass runs under `torch.no_grad()` and produces T_{HQ} and T_{feat}; the harder Student pass produces S_{HQ} and S_{feat}. Both passes share weights, and only the Student pass receives gradients.

The training objective combines reconstruction, perceptual, adversarial, and consistency losses:

$$L_{total} = w_{pix}L_{pix} + w_{per}L_{per} + w_{GAN}L_{GAN} \; + \lambda_{kd}^{out}L_{kd}^{out} + \lambda_{kd}^{feat}L_{kd}^{feat}.$$

L_{pix} is L1 loss to GT, L_{per} is VGG19 perceptual loss [?,?], and L_{GAN} uses a spectral-normalized U-Net discriminator. L_{kd}^{out} and L_{kd}^{feat} align Student outputs and features with detached Teacher targets, giving dynamic guidance without a second stored network.

3.3 Curriculum Schedule

The curriculum schedule exposes the model to different degradation severities in a controlled mild-to-hard order, instead of training on all levels randomly from the beginning. Following curriculum learning [?], we define training difficulty by degradation strength. In our setting, 100% denotes the original GT image, while lower percentages indicate stronger degradation. For each HR patch, the degradation pipeline generates LQ versions at 90%, 80%, 70%, and 60%, which are paired according to the current training phase.

Training is divided into three phases. *Phase 1* uses the 90%/80% pair, where both inputs are mildly degraded and most structural information is preserved, providing a stable warm-up. *Phase 2* uses the 80%/70% pair, increasing the degradation severity and encouraging the model to recover more corrupted textures and edges. *Phase 3* uses the 70%/60% pair, where the harder input contains stronger blur, noise, compression, and camera-related artifacts, improving robustness to severe degradation.

In each phase, the first element of the pair serves as the milder guidance input, while the second element is used as the harder restoration input. Since both images are generated from the same HR patch, they remain content-aligned and can support output-level and feature-level consistency learning. Through this progressive schedule, the model gradually adapts to increasingly difficult restoration scenarios without being exposed to the strongest degradations too early.

4 Experiments

4.1 Experimental Setup

Training uses high-resolution images from DF2K and Flickr2K [**?**], Unsplash [**?**], and RealSR [**?**]. Only HQ images are retained; all LQ images are generated by our degradation pipeline. This better approximates real acquisition but makes pixel-faithful reconstruction harder. Evaluation uses Set5 [**?**], Set14 [**?**], RealSR Canon/Nikon [**?**], BSD100 [**?**], and Urban100 [**?**].

We report PSNR, SSIM, and NIQE [**?**]; lower NIQE indicates better perceptual naturalness in blind SR.

All models are trained with Adam using an initial learning rate of 1×10^{-4} for 800k iterations, including a 10k-iteration warm-up. MultiStepLR decays the learning rate at 200k, 400k, 500k, and 700k iterations. Loss weights are set to $w_{pix} = 2.0$, $w_{GAN} = 0.5$, and $\lambda_{kd}^{out} = \lambda_{kd}^{feat} = 0.15$. The 16.7M-parameter generator uses a single inference pass, so the proposed training scheme adds no inference-time model size or runtime complexity. Experiments are conducted on an NVIDIA RTX 3090 GPU.

4.2 Result Analysis

Table 1 focuses on Real-ESRGAN variants to make the comparison fair and attributable. The first two rows report the performance of the original Real-ESRGAN [**?**] with the original and multi-level settings, while the last three rows show performance of our framework at different phases. Here, the multi-level setting refers to our generated degradation levels, where each HR image is converted into controlled 90%, 80%, 70%, and 60% LQ versions for paired curriculum training. The *Real-ESRGAN with Multi-level Degradation* row isolates the degradation-setting effect without Siamese training: it improves PSNR/SSIM on RealSR Canon/Nikon and BSDS100, but high NIQE shows that richer degradation alone does not guarantee perceptually natural restoration. With Siamese curriculum learning, Phase 2 reaches NIQE 3.82 on Urban100 and reduces RealSR_V3 Canon NIQE from 7.02 to 5.71, an 18.7% gain. Phase 3 further improves NIQE on RealSR Canon/Nikon and Set14, at the cost of lower PSNR/SSIM due to stronger degradation.

Table 1. Comparison of Real-ESRGAN variants on benchmark datasets. The original row uses the released Real-ESRGAN model with online high-order degradation; all other rows use our generated multi-level degradation setting.

Model	Degradation Setting	Set5		Set14		RealSR_V3 (Canon)		RealSR_V3 (Nikon)		BSDS100		Urban100		Avg. Time (s)
		PSNR/SSIM	NIQE	PSNR/SSIM	NIQE	PSNR/SSIM	NIQE	PSNR/SSIM	NIQE	PSNR/SSIM	NIQE	PSNR/SSIM	NIQE	
Real-ESRGAN[**?**]	Original online	**26.31/0.80**	7.89	**25.18/0.69**	5.88	25.84/0.78	7.02	25.19/0.75	6.69	22.34/0.57	4.46	20.12/**0.64**	4.35	1.5
Real-ESRGAN[**?**]	Multi-level	24.40/0.67	10.44	24.52/0.67	6.10	**27.55/0.81**	7.05	**26.24/0.77**	6.94	**22.49/0.58**	4.83	20.06/0.61	4.33	1.5
Real-ESRGAN student[**?**]	Multi-level	25.19/0.76	**6.54**	23.89/0.69	5.82	26.77/0.80	6.78	25.93/0.76	6.25	22.31/0.57	4.82	20.00/0.59	4.05	1.45
Our framework - Phase1	80%/70%	25.42/0.79	8.52	24.51/0.68	6.69	26.35/0.78	6.37	25.60/0.74	6.13	21.45/0.60	5.18	**21.07**/0.64	3.90	1.72
Our framework - Phase 2	80%/70%	25.37/0.71	7.98	24.69/0.64	5.91	25.21/0.74	5.71	24.76/0.71	5.69	22.45/**0.57**	**4.25**	20.57/0.58	**3.82**	1.72
Our framework - Phase 3	70%/60%	25.98/0.65	8.00	23.44/0.59	**5.58**	24.76/0.73	**5.64**	23.28/0.69	**5.47**	22.70/0.54	4.28	19.80/0.55	3.93	**1.37**

The results follow the perception-distortion trade-off [**?**]: PSNR/SSIM favor pixel alignment and simpler degradations, whereas NIQE reflects naturalness in blind SR.

Our multi-level degradation setting can reduce PSNR/SSIM on synthetic benchmarks, but improves NIQE when degradation is closer to real camera pipelines.

Fig. **??** compares the visual outputs of different models. Phase 2 shows sharper text, cleaner edges, less aliasing on Urban100, and fewer texture artifacts on BSDS100, consistent with the NIQE trends in Table 1.

4.3 Ablation Study

Table 1 analyzes the effects of degradation design, Siamese guidance, and curriculum severity. Comparing the released Real-ESRGAN with the backbone trained using our multi-level degradation setting shows that richer degradation improves PSNR/SSIM on RealSR_V3 Canon/Nikon and BSDS100, but does not alone ensure perceptually natural results, as indicated by the high NIQE values. Adding the shared-weight Siamese guidance improves perceptual quality: Phase 2 achieves the best Urban100 NIQE and improves RealSR_V3 Canon NIQE over the baseline. Comparing Phase 2 and Phase 3 further shows that Phase 2 provides the best overall perceptual balance, while Phase 3 is more oriented toward robustness under the strongest degradation pair.

We also examined implementation variants. Training directly on only the hardest 70%/60% pair for 120k iterations achieved 22.13 dB PSNR and 0.665 SSIM on Set5, with NIQE scores of 6.26 on RealSR Canon, 6.22 on RealSR Nikon, and 4.14 on Urban100. These results are worse than the curriculum-trained Phase 2/Phase 3 variants, confirming the benefit of gradual 90%/80% to 70%/60% progression. Using a separate fixed Teacher increased memory without improving NIQE, while disabling feature-level KD reduced texture consistency. Overall, the results support the proposed design: multi-level degradation provides diverse paired inputs, Siamese guidance improves perceptual restoration, and curriculum learning controls degradation difficulty.

5 Conclusion

Pairing two degradation severities of the same image through a shared-weight Siamese training strategy consistently improves perceptual quality without increasing inference cost. The proposed curriculum achieves the best NIQE of 3.82 on Urban100 and reduces NIQE on RealSR_V3 Canon from 7.02 to 5.71 (18.7%) compared with the original Real-ESRGAN. These results suggest that progressively increasing degradation difficulty is more effective than directly training on the most challenging samples. Future work will investigate automatic curriculum scheduling and lightweight generative refinement to further improve restoration under severe real-world degradations.

References

1. Bengio, Y., Louradour, J., Collobert, R., Weston, J.: Curriculum learning. In: Proceedings of the 26th Annual International Conference on Machine Learning. p. 41–48. ICML ’09, Association for Computing Machinery, New York, NY, USA (2009). https://doi.org/10.1145/1553374.1553380, https://doi.org/10.1145/1553374.1553380

2. Bevilacqua, M., Roumy, A., Guillemot, C., Alberi-Morel, M.L.: Low-complexity single image super-resolution based on nonnegative neighbor embedding (09 2012). https://doi.org/10.5244/C.26.135
3. Blau, Y., Michaeli, T.: The perception-distortion tradeoff. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 6228–6237 (2018). https://doi.org/10.1109/CVPR.2018.00652
4. Cai, J., Zeng, H., Yong, H., Cao, Z., Zhang, L.: Toward real-world single image super-resolution: A new benchmark and a new model (2019), https://arxiv.org/abs/1904.00523
5. Dong, C., Loy, C.C., He, K., Tang, X.: Image super-resolution using deep convolutional networks (2015), https://arxiv.org/abs/1501.00092
6. Hinton, G., Vinyals, O., Dean, J.: Distilling the knowledge in a neural network (2015), https://arxiv.org/abs/1503.02531
7. Huang, J.B., Singh, A., Ahuja, N.: Single image super-resolution from transformed self-exemplars (06 2015). https://doi.org/10.1109/CVPR.2015.7299156
8. Johnson, J., Alahi, A., Fei-Fei, L.: Perceptual losses for real-time style transfer and super-resolution (2016), https://arxiv.org/abs/1603.08155
9. Liang, J., Cao, J., Sun, G., Zhang, K., Gool, L.V., Timofte, R.: Swinir: Image restoration using swin transformer (2021), https://arxiv.org/abs/2108.10257
10. Lim, B., Son, S., Kim, H., Nah, S., Lee, K.M.: Enhanced deep residual networks for single image super-resolution (2017), https://arxiv.org/abs/1707.02921
11. Martin, D.R., Fowlkes, C.C., Tal, D., Malik, J.: A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. Proceedings Eighth IEEE International Conference on Computer Vision. ICCV 2001 **2**, 416–423 vol.2 (2001), https://api.semanticscholar.org/CorpusID:64193
12. Mittal, A., Soundararajan, R., Bovik, A.C.: Making a "completely blind" image quality analyzer. IEEE Signal Processing Letters **20**(3), 209–212 (2013). https://doi.org/10.1109/LSP.2012.2227726
13. Nam, L.H., Quang, D.N., Tuan, G.A., Tung, T.H., Ton, L.H., Ha, N.H.: Towards real-esrgan refinement for realistic image restoration via advanced degradation and knowledge distillation. In: 2025 17th International Conference on Knowledge and System Engineering (KSE). pp. 01–06 (2025). https://doi.org/10.1109/KSE68178.2025.11309534
14. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition (2015), https://arxiv.org/abs/1409.1556
15. Timofte, R., Agustsson, E., Van Gool, L., Yang, M.H., Zhang, L., Lim, B., Son, S., Kim, H., Nah, S., Lee, K.M., Wang, X., Tian, Y., Yu, K., Zhang, Y., Wu, S., Dong, C., Lin, L., Qiao, Y., Loy, C.C., Guo, Q.: Ntire 2017 challenge on single image super-resolution: Methods and results. pp. 1110–1121 (07 2017). https://doi.org/10.1109/CVPRW.2017.149
16. Unsplash: Unsplash dataset. https://unsplash.com/data (2026)
17. Wang, X., Xie, L., Dong, C., Shan, Y.: Real-esrgan: Training real-world blind super-resolution with pure synthetic data (2021), https://arxiv.org/abs/2107.10833
18. Wang, X., Yu, K., Wu, S., Gu, J., Liu, Y., Dong, C., Loy, C.C., Qiao, Y., Tang, X.: Esrgan: Enhanced super-resolution generative adversarial networks (2018), https://arxiv.org/abs/1809.00219
19. Zeyde, R., Elad, M., Protter, M.: On single image scale-up using sparse-representations. In: Curves and Surfaces (2010), https://api.semanticscholar.org/CorpusID:2356330
20. Zhang, K., Liang, J., Gool, L.V., Timofte, R.: Designing a practical degradation model for deep blind image super-resolution (2021), https://arxiv.org/abs/2103.14006
21. Zhang, Y., Tian, Y., Kong, Y., Zhong, B., Fu, Y.: Residual dense network for image super-resolution (2018), https://arxiv.org/abs/1802.08797